

Coordinate geometry

a) Parallel and perpendicular line

Example 1

P has coordinates $(-5, 4)$ and Q has coordinates $(-3, -1)$.

Find

- the gradient of PQ
- the length of PQ
- the coordinates of the mid-point of PQ .

$$\begin{aligned} \text{a) gradient} &= \frac{-1-4}{-3-(-5)} \\ &= \frac{-5}{2} = -\frac{5}{2} \end{aligned}$$

If you start with -1 in the numerator you must start with -3 in the denominator.

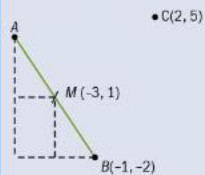
$$\begin{aligned} \text{b) length } PQ &= \sqrt{(-3-(-5))^2 + (-1-4)^2} \\ &= \sqrt{(2)^2 + (-5)^2} \\ &= \sqrt{29} = 5.385 \end{aligned}$$

$$\begin{aligned} \text{c) mid-point of } PQ &= \left(\frac{-5+(-3)}{2}, \frac{4+(-1)}{2} \right) \\ &= (-4, 1.5) \end{aligned}$$

Check: -4 is halfway between -3 and -5 .
 1.5 is halfway between 4 and -1 .

Example 2

The mid-point M of the line AB has coordinates $(-3, 1)$. B has coordinates $(-1, -2)$.
Given that a point C has coordinates $(2, 5)$, find the exact length of the line AC .



It is useful to draw a diagram.

Let A be the point (x, y) .

Work out the coordinates of A .

$$\text{Then } \frac{x+(-1)}{2} = -3, \text{ so } x = -5$$

$$\text{Also } \frac{y+(-2)}{2} = 1, \text{ so } y = 4$$

$$A = (-5, 4) \quad C = (2, 5)$$

$$\begin{aligned} \text{length } AC &= \sqrt{(2-(-5))^2 + (5-4)^2} \\ &= \sqrt{1+49} = \sqrt{50} \end{aligned}$$

We are asked for an *exact* answer so do not work out answer as a decimal.

Example 5

Find the gradient of the line that is perpendicular to the lines with these gradients.

Note: Since $m_1 \times m_2 = -1$, $m_2 = -\frac{1}{m_1}$

- a) $\frac{7}{3}$ b) $-\frac{1}{5}$ c) 1 d) -8

a) $-\frac{3}{7}$

Change the sign and invert the fraction.

b) $\frac{5}{1} = 5$

Change the sign and invert the fraction.

c) $-\frac{1}{1} = -1$

$1 = \frac{1}{1}$ and $-8 = -\frac{8}{1}$

d) $\frac{1}{8}$

Example 6

Find in each case whether the lines are parallel to each other, perpendicular to each other, or neither.

- a) $y = 5 - 3x$ b) $x - 2y = 4$ c) $3y = 5x - 2$
 $y = 3x + 4$ $6y = 3x - 1$ $10y = 3 - 6x$

a) $m_1 = -3$ $m_2 = 3$

The lines are not parallel or perpendicular.

The gradient is equal to the coefficient of x .

b) $m_1 = \frac{1}{2}$ $m_2 = \frac{3}{6} = \frac{1}{2}$

The lines are parallel, as $m_1 = m_2$.

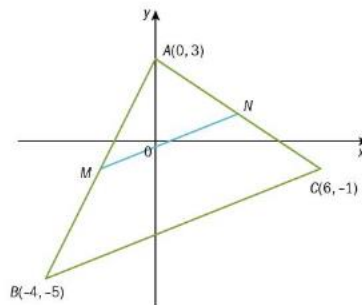
$2y = x - 4$, $y = \frac{1}{2}x - 2$ and $y = \frac{3}{6}x - \frac{1}{6}$

c) $m_1 = \frac{5}{3}$ $m_2 = -\frac{6}{10} = -\frac{3}{5}$

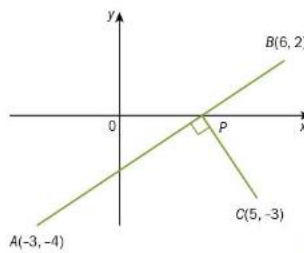
The lines are perpendicular, as $m_1 m_2 = -1$.

$y = \frac{5}{3}x - \frac{2}{3}$ and $y = \frac{3}{10} - \frac{6}{10}x$

3. Find in each case whether the lines are parallel to each other, perpendicular to each other, or neither.
- a) $y = 1 - x$ b) $x - 2y = 4$
 $y = x + 4$ $6y = 3x - 1$
- c) $3y = 9x + 1$ d) $4y = 8x + 1$
 $x + 3y = 4$ $2y = 3 - 4x$
4. The line PQ has a gradient of $\frac{1}{3}$. PQ is parallel to the line joining the points $(3, k)$ and $(-6, 5)$.
Find k .
5. The line joining points $A(x, 3)$ and $B(2, -3)$ is perpendicular to BC where C is the point $(10, 1)$.
Find the value of x .
6. The line that passes through the points $(3, 0)$ and $(-4, y)$ is perpendicular to the line with equation $7x + 4y = 5$. Find the value of y .
7. ABC is a right-angled triangle with AB perpendicular to BC . The coordinates of the points A, B and C are $(-5, 0)$, $(x, -5)$ and $(3, -2)$ respectively. Find the possible values of x .
8. Show that $A(4, 3)$, $B(-3, 2)$, $C(-5, -2)$ and $D(2, -1)$ are the vertices of a parallelogram.
9. The diagram shows triangle ABC . The coordinates of A, B and C are $(0, 3)$, $(-4, -5)$ and $(6, -1)$ respectively. M is the mid-point of AB , and N is the mid-point of AC .
Show that MN is parallel to BC .



10. The diagram shows the straight line AB that cuts the x -axis at P . PC is perpendicular to AB . The point A is $(-3, -4)$, the point B is $(6, 2)$ and the point C is $(5, -3)$.
Show that point P lies on the x -axis and find its coordinates.



b) Equation of straight line

Example 8

Find the equation of the line that passes through the point $(4, -3)$ and is perpendicular to the line with equation $2y = 5x$.

$$2y = 5x \quad m = \frac{5}{2}$$

$$\text{Gradient of the perpendicular line} = -\frac{2}{5}$$

$$\text{Equation: } y + 3 = -\frac{2}{5}(x - 4)$$

$$5y + 15 = -2x + 8$$

$$2x + 5y + 7 = 0$$

← The gradient of the line is equal to the coefficient of x .

← Change the sign and inverse the fraction.

← Multiply $(y + 3)$ by 5 and multiply $(x - 4)$ by -2 .

Example 9

Find the equation of the **perpendicular bisector** of the line PQ , where $P(3, -1)$ and $Q(7, 3)$.

$$\text{Mid-point of } PQ = \left(\frac{3+7}{2}, \frac{-1+3}{2} \right) = (5, 1)$$

$$\text{Gradient } PQ = \frac{3 - (-1)}{7 - 3} = 1$$

$$\text{Gradient of perpendicular} = -1$$

$$\text{Equation of perpendicular:}$$

$$y - 1 = -1(x - 5)$$

$$x + y - 6 = 0$$

$$m_1 \times m_2 = -1$$

Example 10

$A(4, 3)$, $B(9, 13)$, $C(5, 11)$, and $D(12, 4)$ are four points.

The lines AB and CD meet at P .

Find the equations of AB and CD and hence the coordinates of P .

$$\text{Gradient of } AB = \frac{13-3}{9-4} = 2$$

$$\text{Equation } AB: y - 3 = 2(x - 4)$$

$$y = 2x - 5$$

$$\text{Gradient of } CD = \frac{4-11}{12-5} = -1$$

$$\text{Equation } CD: y - 11 = -1(x - 5)$$

$$y = -x + 16$$

$$\text{At } P: 2x - 5 = -x + 16$$

$$3x = 21$$

$$x = 7$$

$$\text{When } x = 7, y = 2(7) - 5 = 9$$

$$\text{Coordinates of } P = (7, 9)$$

← Use the coordinates of A or B .

← Use the coordinates of C or D .

← Both equal y .

← Use equation of AB or CD .

2. Find the equation of the line AB with the given pairs of coordinates.

- a) $A(1, 5)$ and $B(3, 7)$ b) $A(-1, -2)$ and $B(-3, 4)$
 c) $A(5, -3)$ and $B(-2, 4)$ d) $A(-2, 4)$ and $B(-6, 0)$
 e) $A(-8, -1)$ and $B(2, -6)$ f) $A(9, -3)$ and $B(-6, -2)$

3. Find the equation of the line parallel to the line $y = 4 - 3x$ and passing through the point $(5, 7)$.

4. Find the equation of the line perpendicular to the line $y = 5x$ and passing through the point $(2, -1)$.

5. Find the equation of the line through the point $(-1, 3)$ parallel to the line with equation $2x - 5y = 10$.

6. Find the equation of the line through the point $(-1, 3)$ perpendicular to the line with equation $6x + 9y - 7 = 0$.

7. Find the equation of the line through the point $(1, 8)$ perpendicular to the line with equation $3x - 4y + 4 = 0$.

8. A is the point $(3, -4)$ and B is the point $(-5, -10)$. Find the equation of the perpendicular bisector of AB .

9. $A(9, 0)$, $B(5, 8)$, $C(1, 1)$ and $D(3, 2)$ are four points.

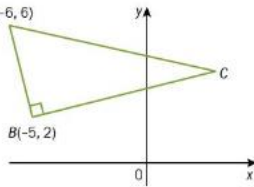
The lines AB and CD meet at P .

Find the equations of AB and CD and hence the coordinates of P .

10. $A(1, -2)$, $B(-1, -3)$, $C(-2, 14)$, and $D(2, 2)$ are four points.

The lines AB and CD meet at P . Find the coordinates of P .

11. $A(-6, 6)$



The diagram shows triangle ABC with AB perpendicular to BC . A is the point $(-6, 6)$ and B is the point $(-5, 2)$. The gradient of AC is $-\frac{2}{9}$. Find the coordinates of C .

12. The straight line with equation $3x - 2y = 6$ cuts the x -axis at P and the y -axis at Q .

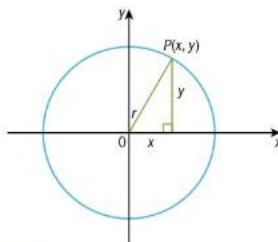
a) Find the coordinates of P and Q .

b) Find the equation of the line that is perpendicular to PQ and passes through the point $(4, 3)$.

13. $A(5, -5)$, $B(3, -7)$, $C(12, 3)$ and $D(-3, -6)$ are four points.

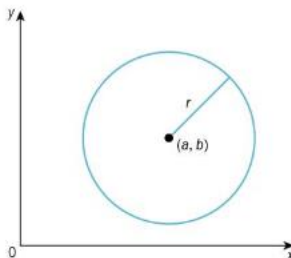
c) Circles

If we take any point $P(x, y)$ on the circumference of a circle, centre O , we can work out the equation of the circle using Pythagoras' theorem.



The equation of a circle, centre $(0, 0)$, radius r , is given by $x^2 + y^2 = r^2$.

If we translate this circle by the vector $\begin{pmatrix} a \\ b \end{pmatrix}$, we can get the equation of any circle, centre (a, b) , radius r .



The equation of a circle, centre (a, b) , radius r is given by $(x - a)^2 + (y - b)^2 = r^2$.

If we expand this equation, we get $x^2 - 2ax + a^2 + y^2 - 2by + b^2 = r^2$. When this is simplified, we get the expanded form of the equation of a circle:

$$x^2 + y^2 - 2ax - 2by + c = 0 \quad \text{where } c = a^2 + b^2 - r^2$$

Note: When the circle is given in this expanded form, we can use the method of completing the square to write the equation in the form $(x - a)^2 + (y - b)^2 = r^2$.

The expanded form of the equation of a circle is given by

$$x^2 + y^2 + 2gx + 2fy + c = 0, \text{ where } c = f^2 + g^2 - r^2.$$

Example 12

Find the centre and radius of the circle with equation $x^2 + y^2 - 10x + 12y + 12 = 0$.

$$x^2 - 10x + y^2 + 12y + 12 = 0$$

$$(x - 5)^2 - 25 + (y + 6)^2 - 36 + 12 = 0$$

$$(x - 5)^2 + (y + 6)^2 = 49$$

Centre is $(5, -6)$, radius = 7.

First rearrange the equation to this form.

Complete the square for x and y .

Rearrange to this form.

$$a = 5, b = -6, r = \sqrt{49}$$

Example 13

Find the equation of the circle with centre $(4, -3)$ that passes through the point $(-2, 5)$.

$$(x - 4)^2 + (y + 3)^2 = r^2$$

$$(-2 - 4)^2 + (5 + 3)^2 = r^2$$

$$36 + 64 = r^2$$

$$r^2 = 100$$

Equation of the circle is

$$(x - 4)^2 + (y + 3)^2 = 100$$

Write the equation of circle centre $(4, -3)$.

Substitute for $x = -2$ and $y = 5$.

Simplify

Work out r^2 .

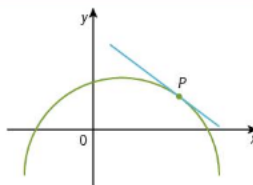
Write down the equation of the circle.

d) Points of intersection

When a line touches a curve in exactly one place we call that line a **tangent** to the curve.

This also means that there will only be one solution when solving the equations simultaneously.

If you form a quadratic equation there will be only **one (repeated) root**,
i.e. $b^2 - 4ac = 0$



Example 14

Find the set of values of k for which the line $x = y + 2$ intersects the curve $y = x^2 + 3x + k$ at two distinct points.

They meet when $x^2 + 3x + k = x - 2$.

$$x^2 + 2x + (k + 2) = 0$$

Rearrange as a quadratic equation.

There are two distinct roots when $b^2 - 4ac > 0$.

$$(2)^2 - 4(1)(k + 2) > 0$$

$a = 1, b = 2$ and $c = k + 2$

$$-4k - 4 > 0$$

$$-4k > 4$$

Use inequality signs throughout.

$$k < -1$$

Example 15

A line has equation $y = 4k - x$ and a curve has equation $y = 3x - kx^2$, where k is a constant.

- a) Find the two values of k for which the line is a tangent to the curve.
b) Hence find the coordinates of the points where the line touches the curve.

- a) The line and the curve meet when

$$4k - x = 3x - kx^2$$

Both equal y .

$$kx^2 - 4x + 4k = 0$$

Rearrange in the form $ax^2 + bx + c = 0$.

The line is a tangent so there is one repeated root.

$$(-4)^2 - 4(k)(4k) = 0$$

$$b^2 - 4ac = 0$$

$$16 = 16k^2, k^2 = 1$$

$$k = 1 \text{ or } k = -1$$

- b) When $k = 1$:

$$x^2 - 4x + 4 = 0$$

Substitute in $kx^2 - 4x + 4k = 0$.

$$(x - 2)(x - 2) = 0$$

$$x = 2, y = 4k - x = 4 - 2 = 2$$

When $k = -1$:

$$-x^2 - 4x - 4 = 0$$

$$x^2 + 4x + 4 = 0$$

$$(x + 2)(x + 2) = 0$$

$$x = -2, y = 4k - x = -4 + 2 = -2$$

Line meets curve at $(2, 2)$ and $(-2, -2)$.

The question asked for coordinates.

Example 16

Determine the equation of the tangent to the circle, centre C , with equation $x^2 + y^2 - 2x - 4y - 5 = 0$ at the point $A(2, 5)$. Give your answer in the form $ax + by + c = 0$, where a , b and c are integers.

$$(x - 1)^2 - 1 + (y - 2)^2 - 4 - 5 = 0$$

Complete the square to find the centre.

C is $(1, 2)$.

We do not need to find the radius.

$$\text{Gradient of } AC = \frac{5-2}{2-1} = 3$$

Use the points $(2, 5)$ and $(1, 2)$.

$$\text{Gradient of the tangent} = -\frac{1}{3}$$

The tangent is perpendicular to the radius.

$$\text{Equation of the tangent: } y - 5 = -\frac{1}{3}(x - 2)$$

The gradient is $-\frac{1}{3}$ through the point $(2, 5)$.

$$3y - 15 = -x + 2$$

$$x + 3y - 17 = 0$$

Simplify and put in the required form.

Example 17

Determine the shortest distance from the point $P(1, 1)$ to the circle, centre C , with equation $x^2 + y^2 + 6x - 8y + 21 = 0$.

$$(x + 3)^2 - 9 + (y - 4)^2 - 16 + 21 = 0$$

Complete the square to find C and radius r .

C is $(-3, 4)$, $r = \sqrt{4} = 2$.

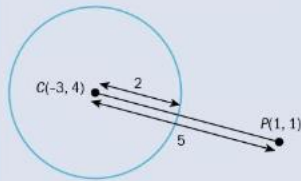
Find the centre and the radius.

$$PC = \sqrt{(1 - -3)^2 + (1 - 4)^2} = 5$$

Find the distance between P and C .

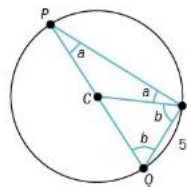
$$\text{Shortest distance} = 5 - 2 = 3$$

See diagram below.



Remember that the tangent to a point on a circle is always perpendicular to the radius at that point and that angles in a semicircle are always right angles. The diagram reminds you why $\angle PRQ = 90^\circ$ if PQ is a diameter of a circle with centre C . Drawing the radius CR creates two isosceles triangles, with one base angle from each making up $\angle PRQ$.

You can be asked to use these elementary geometrical properties of circles when working with coordinate geometry.

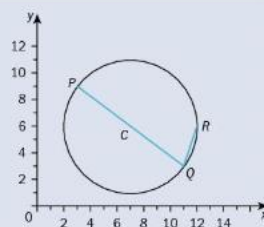


Coordinate geometry

Example 18

The diagram shows a circle, centre C , with equation $(x - 7)^2 + (y - 6)^2 = 25$. PQ is a diameter, where P is $(3, 9)$ and Q is $(11, 3)$. $R(12, 6)$ lies on the circle.

Show that the line perpendicular to QR and passing through R goes through P .



QR has gradient $\frac{6-3}{12-11} = 3$, so the gradient

of the perpendicular is $-\frac{1}{3}$.

Gradients of perpendicular lines multiply to -1 .

The equation of the perpendicular line through $(12, 6)$ is given by

$$y - 6 = -\frac{1}{3}(x - 12)$$

Substitute the gradient and known point to get the equation of the line.

$$y = -\frac{1}{3}x + 10$$

Rearrange to standard form.

When $x = 3$, $y = -\frac{1}{3} \times 3 + 10 = 9$

so $P(3, 9)$ lies on the line.

Always finish the answer to a 'show that' question with a statement.

Example 19

AB is the diameter of a circle, where A is $(2, 6)$ and B is $(8, 2)$.

The tangent to the circle at B meets the x -axis at P and the y -axis at Q . Find the coordinates of P and Q .

A tangent is perpendicular to the diameter and we have two points defining the diameter.

AB has gradient $\frac{2-6}{8-2} = -\frac{2}{3}$, so the

gradient of the tangent at B is $\frac{3}{2}$.

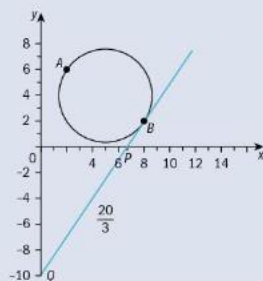
The equation of the perpendicular line through $(8, 2)$ is given by

$$y - 2 = \frac{3}{2}(x - 8)$$

Substitute the gradient and known point to get the equation of the line.

$$y = \frac{3}{2}x - 10$$

Rearrange to standard form.



Continued on the next page

When $y = 0$, $0 = \frac{3}{2}x - 10 \Rightarrow x = \frac{20}{3}$ ← Substitute $x = 0$ and $y = 0$ to find coordinates of intercepts.

so P is $\left(\frac{20}{3}, 0\right)$.

When $x = 0$, $y = -10$ so Q is $(0, -10)$.

Example 20

The points P , Q and R have coordinates $(0, 2)$, $(4, 0)$ and $(8, 8)$ respectively.

Use Pythagoras' theorem to show that PQR is a right-angled triangle and hence find the equation of the circle that passes through P , Q and R .

$$PQ^2 = 2^2 + 4^2 = 20 \quad QR^2 = 4^2 + 8^2 = 80$$

$$PR^2 = 8^2 + 6^2 = 100$$

so $PQ^2 + QR^2 = PR^2$ and PQR is a right-angled triangle with PR as the hypotenuse with the right angle at Q .

The circle through P , Q and R therefore has PR as diameter.

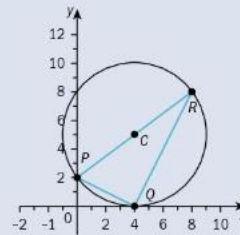
If C is the centre of the circle, then C is the mid-point of PR and the radius of the circle is half PR .

C is $(4, 5)$, radius = 5.

The equation of the circle is $(x - 4)^2 + (y - 5)^2 = 25$.

Calculate the squares of sides of triangle.

Use Pythagoras' theorem as required.



Substitute centre coordinates and radius to get the equation of the circle.

Summary:

Mid-point of a line segment

- The mid-point of the line joining (x_1, y_1) and (x_2, y_2) is given by $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$.

Gradient of a line

- The gradient of the line joining (x_1, y_1) and (x_2, y_2) is given by $\frac{y_2 - y_1}{x_2 - x_1}$.

Length of a line segment

- The length of the line segment between (x_1, y_1) and (x_2, y_2) is given by $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Parallel and perpendicular lines

- Two straight lines are parallel if their gradients are the same ($m_1 = m_2$).
- Two straight lines are perpendicular if the product of their gradients = -1 ($m_1 \times m_2 = -1$).

Equation of a line

- The equation of the straight line through the point (x_1, y_1) and with gradient m is:
 $(y - y_1) = m(x - x_1)$ (given point and gradient).
- The equation of the straight line through the points (x_1, y_1) and (x_2, y_2) is:
 $\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$ (given two points).

Equation of a circle

- The equation of a circle, centre (a, b) , radius r , is given by $(x - a)^2 + (y - b)^2 = r^2$.
- The expanded form of the equation of a circle is given by $x^2 + y^2 + 2gx + 2fy + c = 0$,
where $c = f^2 + g^2 - r^2$.

Tangent to a curve

- When a line touches a curve in exactly one place we call that line a tangent to the curve.
- When you form a quadratic equation in order to solve the equation of the tangent and the equation of the quadratic simultaneously, there will be only one repeated root, i.e. use $b^2 - 4ac = 0$.